Iyed MEDIOUNI

AI ENGINEER · LLM SYSTEMS & APPLIED MACHINE LEARNING

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PROFILE

AI engineer who takes language-model systems all the way into production. I build assistants that answer from a company's own live systems, and the evaluation loop that proves each version is actually better before it ships. The assistant I built during my internship was sold to a client. Comfortable across the whole path: retrieval design, multi-agent orchestration, on-premise inference, FastAPI services and the console the team runs it from.

EXPERIENCE

02/2026 - 07/2026
Lac 1, Tunis
Final-year project

AI Engineer - TELNET Holding · AI assistant with fine-tuning ● github.com/iyed122/ai-assistant

- Project knowledge sat fragmented across Jira, Confluence and GitLab: delivered a production, on-premise assistant on LangGraph and Neo4j Graph-RAG that answers across all three from one natural-language interface. TELNET sold the product to a client.
- Scaled to the client corpus: 1.28 M segments embedded at 768 dimensions (int8-quantised HNSW index, sized to a single 6 GB card) across 428,593 tickets and 47,851 pages, linked by 1.8 M+ typed relations - hybrid vector search with multi-hop graph traversal.
- Responses were too slow to feel interactive: built a hybrid query-routing layer pairing deterministic classification with parallel API dispatch, cutting mean latency 2.4x (7.3s to 3.0s).
- Designed Hammer, an evaluation framework (RAGAS + deterministic validators) grading every answer on two axes; its deterministic veto keeps detected fabrications out of training - 62 kept out of a 138-document candidate pool.
- RAFT fine-tuning of Qwen3-8B (4-bit QLoRA) on Hammer's verdicts: grounding in a verifiable citation rose from 0 to 95.8% and refusals fell from 19.0% to 7.6%. The adapter was promoted to production by a statistical gate fixed before training: 2.1 standard errors on 366 held-out questions, grounding held at 94.8%.
- Took it from prototype to operable: Docker-containerised FastAPI services, MongoDB persistence, local inference via Ollama, MLflow tracking every fine-tuning run, and a React/Vite console for training mode, training-set curation and RAFT review - with the evaluation loop doubling as production monitoring for answer quality, latency and failure type.

PROJECTS

09/2025 - 11/2025

Multi-Agent Content Automation ● github.com/iyed122/automated-twitter-content

- Generic model output drifts from what is actually trending: built a CrewAI + LLaMA-3.1-8B pipeline where specialised agents research live web, search and trend data before any drafting begins.
- Closed the loop to publication through the X API with secure OAuth, plus a grounding step tying every generated post back to its retrieved sources.

07/2025 - 10/2025

Autonomous Vehicle Perception & Trajectory Prediction ● github.com/iyed122/autonomous-vehicule-VLM

- Camera detection alone cannot anticipate where an object is heading: built an end-to-end pipeline on AWS SageMaker pairing a fine-tuned YOLOv8 detector with a custom bird's-eye-view model for LiDAR trajectory prediction, trained in PyTorch to ADE 0.8 / FDE 1.2.
- Made the output legible to non-specialists through a Streamlit dashboard surfacing live detection, trajectories and a CLEAR / CAUTION / BRAKE collision-risk signal across 6-camera and LiDAR views.
- Closed the loop to the driver: a local vision-language model (Qwen2.5-VL) reads the full sensor suite and a text-to-speech layer speaks the verdict aloud - keep going, take a break, or a vehicle is close - so nobody has to watch a screen to stay safe.

02/2023 - 06/2023
Ghazela, Technopole
Final-year project

Full-Stack & AI Developer - AURAX · E-commerce platform development with AI

- Built the complete customer journey in React and Node.js (MERN): search and filters, cart, secure online payment via Stripe, promotion codes, order confirmation, tracking and history.
- Trained a visual recommender (VGG16 to PCA to KNN) over 50,000+ product images, served through a Flask API and working from a user's first visit with no history to learn from.
- Handed the business its own controls: a self-service back-office for catalogue, pricing, promotions and stock with a revenue dashboard, delivered inside the company on GitLab and Linear.

EDUCATION

2023 - 2026	**Engineering Degree - Data Science & Artificial Intelligence** TEK-UP University
2020 - 2023	**Bachelor in Computer Science** ISAMM, Manouba

TECHNICAL SKILLS

AI & ML	LangGraph · LangChain · CrewAI · Graph-RAG · RAGAS · RAFT fine-tuning · Ollama · PyTorch · TensorFlow · Scikit-learn · OpenCV
LANGUAGES	Python · JavaScript · SQL · PL/SQL · R · MATLAB
DATA & BACKEND	Neo4j · Qdrant · MongoDB · PostgreSQL · MySQL · Cassandra · FastAPI · Pandas · NumPy
CLOUD & MLOPS	AWS SageMaker · EC2 · Lambda · S3 · DynamoDB · Redshift · RDS · Docker · MLflow
FRONTEND & TOOLING	React · Vite · Node.js · Streamlit · Git / GitLab · LaTeX

LANGUAGES & CERTIFICATIONS

ARABIC	Native
FRENCH	Professional working proficiency - academic and technical writing
ENGLISH	IELTS Academic, British Council - B2 (6.5 / 9)